

Agustín Roca

PhD Student in Computer Vision & AI



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Buenos Aires, Argentina

EDUCATION

Applied Sciences PhD student - Universidad de San Andrés (UdeSA)

10/2024 - 12/2029

Software Engineer - Instituto Tecnológico de Buenos Aires (ITBA)

03/2017 - 12/2022

WORK EXPERIENCE

Research Assistant - AI and Robotics Laboratory (Universidad de San Andrés)

Buenos Aires, Argentina

Wildlife monitoring system

02/2023 - Present

- Developed YOLOv8-based detector for marsh deer in UAV imagery, achieving mAP@10 of 0.950 (ARGENCON 2024)
- Led comparative study of YOLO and RT-DETR transformer architectures for small-object aerial detection (ICUAS 2025)
- Models surpassed trained human annotators, identifying previously unlabeled animals

Buenos Aires autonomous driving dataset

- Integrated, synchronized & calibrated sensors mounted on the car such as LiDAR, cameras, IMU, GPS & microphone
- Designed & developed the ROS software stack needed for the recording of the dataset
- Recorded the first publicly available Latin American autonomous driving dataset
- Currently curating & testing benchmarks in the recorded dataset

Software Engineer Intern - J.P. Morgan Chase

Buenos Aires, Argentina

- Fully developed and tested software to create efficient solutions to company workflows

11/2021 - 04/2022

PUBLICATIONS

- **A. Roca**, G. Castro, G. Torre, L. Colombo, I. Mas, J. Pereira & J.I. Giribet

"Detection of Endangered Deer Species Using UAV Imagery: A Comparative Study Between Efficient Deep Learning Approaches"

ICUAS 2025. <https://ieeexplore.ieee.org/abstract/document/11007886>

- **A. Roca**, G. Torre, J. Giribet, G. Castro, L. Colombo, I. Mas & J. Pereira

"Efficient Endangered Deer Species Monitoring with UAV Aerial Imagery and Deep Learning"

IEEE ARGENCON 2024. <https://ieeexplore.ieee.org/abstract/document/10735858>

- J. Giribet, I. Mas, **A. Roca**, G. Marzik, G. Torre, C. Rosito & G. Castro

"Base de Datos para Conducción Autónoma. Sensores y Sincronización"

Presented at Jornadas Argentinas de Robótica 2024

Dataset published at CONICET institutional repository: <https://ri.conicet.gov.ar/handle/11336/242774>

- **A. Roca** & N. I. Britos

"Controlling Face's Frame generation in StyleGAN's latent space operations"

ITBA. Engineering Thesis. 2022. <https://ri.itba.edu.ar/handle/123456789/4116>

CONFERENCES

- **KHIPU 2025** - Latin American Meeting in Artificial Intelligence (Poster presented)

- **UdeSA's SCIAA 2025** - Applied Artificial Intelligence Scientific Symposium (Poster presented)

- **UdeSA's SCIAA 2024** - Applied Artificial Intelligence Scientific Symposium (Poster presented)

- **JAR 2024** - Argentinean Robotics Conference (Oral presentation given)

TEACHING

Teaching Assistant - Universidad de San Andrés

Buenos Aires, Argentina

08/2025 - Present

- Co-designed & currently co-teaching "Advanced Computer Vision" (newly created course): Interpretability, continual learning, transfer learning, meta-learning, GANs, normalizing flows, and diffusion models
- Currently co-teaching "Computer Vision": Classical image processing, 3D reconstruction, and deep learning for images, including object detection and segmentation.

Head Teaching Assistant - Universidad de San Andrés

Buenos Aires, Argentina

02/2023 - Present

- Currently coordinating & teaching "Computational thinking" practical classes: Introductory course to programming using Python

LANGUAGES

Spanish - Native Proficiency

English - Bilingual Proficiency - IELTS score: 7.0/9.0